PROJECT ANALYSIS PLAN & TECHNICAL MASTER PLAN

RenoRisk Validation Across Multiple CKD Indications In Human Kidney Tissue Biopsies Using LODO Machine Learning

A Multi-Cohort Integration Framework for Evaluating C4BPB, COL13A1, HABP2, CFH, and CFI Across XX Public GEO Datasets

Project Lead: Amela Jusic, PhD, MSc
Role / Group: Translational Research Lead, Quant Biomarkers
Location: Basel, Switzerland

Execution Lead: Stojanko Tudjarski
Execution Role: Data Analysis Execution Specialist

Date & Version: July 2026 | Version 2.0 (Final Analysis Plan)

Project Overview & Governance: Executive Summary: This document outlines the project governance structure, task allocations, and technical workflow for analyzing the RenoRisk 5-biomarker panel (C4BPB, COL13A1, HABP2, CFH, CFI). Under the strategic direction of Dr. Amela Jusic (Project & Translational Research Lead), Stojanko Tudjarski will execute data ingestion, cross-cohort batch harmonization, machine learning modeling, and reporting across 32 public transcriptomic datasets.

1. Project Roles and Task Allocations

To ensure rigorous translational oversight and reproducible computational execution, project responsibilities are structured across governance, execution, and collaborative validation tracks:

Team Member	Project Title & Role	Primary Responsibilities & Key Deliverables
Amela Jusic, PhD, MSc	Project Lead & Translational Research Lead (Quantitative Biomarkers, Basel)	• Study design and strategic oversight (C4BPB, COL13A1, HABP2, CFH, CFI) • Translational alignment with clinical endpoints (eGFR, %TIF, CKD stages) • Data interpretation and experimental & analytical review of meta-analysis results • Lead author on executive reports and scientific dissemination
Stojanko Tudjarski	Lead Execution Computational Biologist	• Processing and normalization of 32 GEO datasets (Microarray, RNA-Seq, snRNA-Seq) • Cross-cohort harmonization using ComBat-seq and limma • Implementation of LODO cross-validation ML models & ElasticNet regression • Code generation, pipeline reproducibility, and figure preparation

2. Target RenoRisk Biomarker Panel Specification

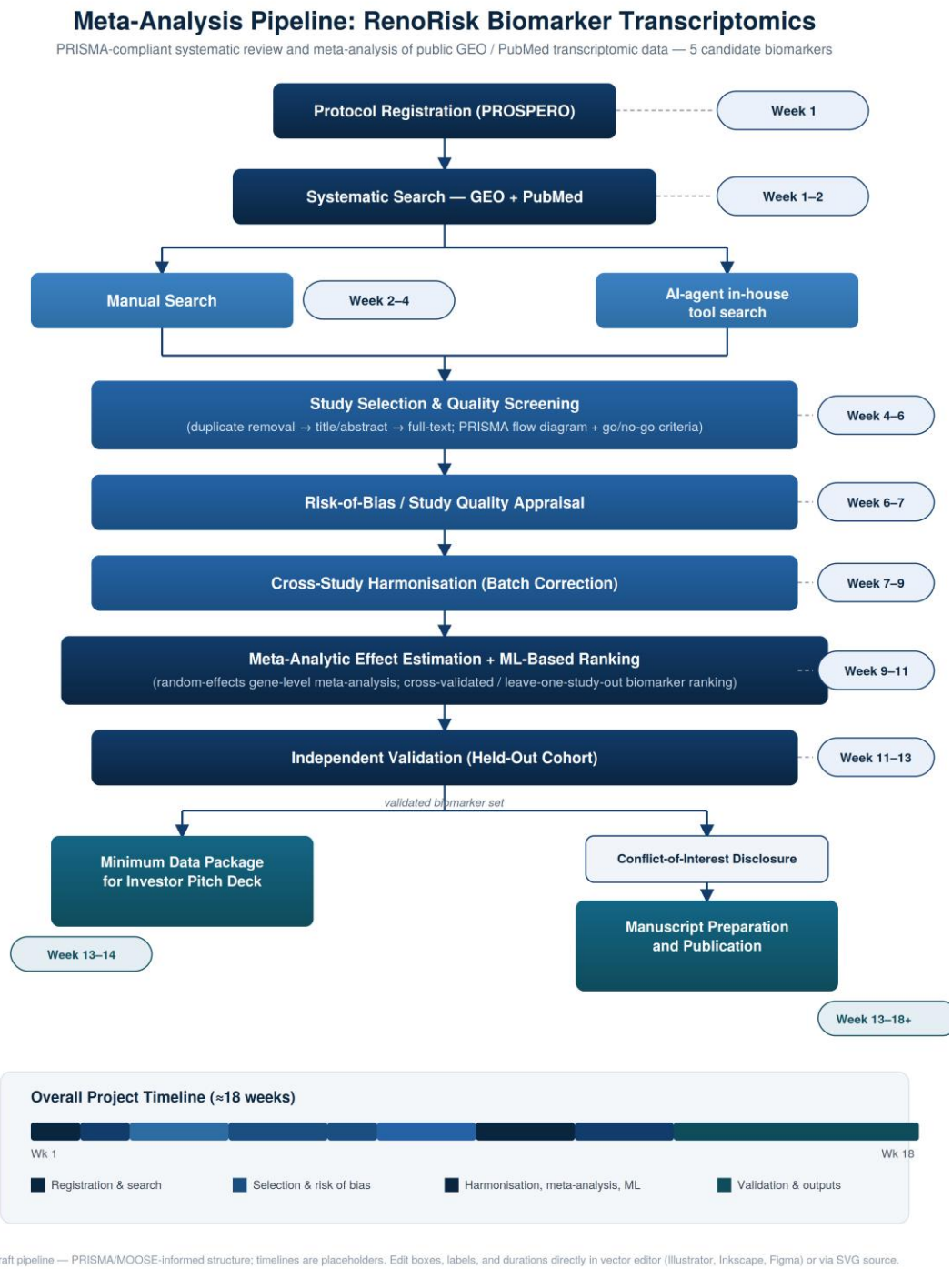
The RenoRisk panel consists of 5 specific target biomarkers categorized into primary diagnostic candidates and additional complement system regulators:

Panel Group	Gene Symbol	Full Gene Name	Biological Role in Renal Disease	GeneCards Identifier

Panel Group	Gene Symbol	Full Gene Name	Biological Role in Renal Disease	GeneCards Identifier
Main RenoRisk	C4BPB	C4b-Binding Protein Beta	Classical complement pathway inhibitor; marker of endothelial/glomerular inflammation	C4BPB
Main RenoRisk	COL13A1	Collagen Type XIII Alpha 1 Chain	Transmembrane collagen involved in cell-matrix adhesion and tubulointerstitial fibrosis	COL13A1
Main RenoRisk	HABP2	Hyaluronan Binding Protein 2	Extracellular serine protease involved in coagulation and renal vascular/extracellular remodeling	HABP2
Additional RenoRisk	CFH	Complement Factor H	Primary alternative pathway complement regulator; central to glomerular endothelial integrity & C3 glomerulopathy/FSGS	CFH
Additional RenoRisk	CFI	Complement Factor I	Essential serine protease that inactivates C3b and C4b; key regulatory gatekeeper in chronic kidney inflammation	CFI

3. Master Analytical Pipeline & Integration Workflow

To evaluate C4BPB, COL13A1, HABP2, CFH, and CFI across diverse public datasets, the raw count/intensity data must undergo uniform harmonization and batch correction to remove technical artifacts while preserving true biological signal.



Step-by-Step Execution Workflow (Stojanko Tudjarski)

- **Stage 1 —** Raw Data Processing & Uniform Quantification: For RNA-Seq cohorts, align raw reads to GRCh38 and quantify transcripts. For Affymetrix microarrays, perform RMA background correction and quantiles normalization.

- **Stage 2 —** Target Gene Mapping: Specifically extract probe sets/transcripts corresponding to C4BPB, COL13A1, HABP2, CFH, and CFI. For microarrays with multiple probes per gene, collapse using WGCNA::collapseRows (selecting maximum mean expression).
- **Stage 3 —** Empirical Bayes Batch Correction: Apply ComBat-seq for raw integer RNA-Seq counts and limma::removeBatchEffect / sva::ComBat for normalized continuous expression, modeling ~ StudyID + Condition.
- **Stage 4 —** Multi-Factor Statistical Modeling: Model differential expression of the 5 RenoRisk genes controlling for tissue compartment (glomerular vs. tubulointerstitial) and patient clinical covariates.

4. Cohort Stratification & GEO Dataset Inventory

The 32 datasets identified in the GEO repository search have been mapped to evaluate RenoRisk target performance across diverse clinical indications: see table in annex 1.

5. Methodology: Pooling Strategy for C4BPB, COL13A1, HABP2, CFH, CFI

To synthesize findings across Microarray and RNA-Seq cohorts without platform bias, we apply summary-level random-effects meta-analysis as the primary framework, complemented by mega-analysis within platform tiers.

Random-Effects Meta-Analysis Model

- **Per-Study Effect Size:** For each RenoRisk gene (C4BPB, COL13A1, HABP2, CFH, CFI), compute Hedges' g standardized mean difference (or log2 fold change) and variance in each dataset.
- **Pooled Summary Estimate:** Combine study effect sizes using DerSimonian-Laird random-effects models via the metafor package in R.
- **Heterogeneity Evaluation:** Evaluate Cochran's Q and I^2 statistics to measure biological and technical heterogeneity across cohorts.

6. Machine Learning Framework for RenoRisk Validation

Machine learning models evaluate whether the 5 RenoRisk biomarkers (C4BPB, COL13A1, HABP2, CFH, CFI) collectively form an accurate, generalizable diagnostic and prognostic panel.

Core ML Tasks & LODO Cross-Validation

- **Task 1 —** Binary Classification: Evaluate Logistic Regression, Random Forest, and Support Vector Machines in distinguishing CKD vs. Control.
- **Task 2 —** eGFR & % Fibrosis Regression: Use ElasticNet regression to predict continuous eGFR and % Tubulointerstitial Fibrosis from expression levels of the 5 RenoRisk genes.
- **Validation —** Leave-One-Dataset-Out (LODO) Validation: Train ML models on N - 1 cohorts and test on the held-out study N. Rotate across all datasets to prevent data leakage and prove external validity.

7. Recommended Visualizations for Publication

- **Plot 1 —** PCA / UMAP Plots (Pre- vs. Post-Batch Correction): Demonstrates batch effect removal across GEO cohorts.
- **Plot 2 —** Forest Plots for 5 RenoRisk Genes: Individual study effect sizes and pooled summary diamonds for C4BPB, COL13A1, HABP2, CFH, and CFI.
- **Plot 3 —** ROC-AUC Comparison Curves: ROC performance comparing single RenoRisk genes against the combined 5-gene ML panel across LODO validation folds.

- **Plot 4 —** Indication Stratified Expression Heatmaps: Expression of C4BPB, COL13A1, HABP2, CFH, CFI across CKD, DKD, IgAN, FSGS, AKI, and Healthy Controls.
- **Plot 5 —** eGFR & % Fibrosis Scatter Plots: Correlation between continuous clinical outcomes and expression levels of COL13A1, CFH, CFI, C4BPB, HABP2.

8. Executable Analysis Code Templates

R Code: DESeq2, ComBat-seq, and Meta-Analysis for 5 RenoRisk Genes

```
# R Script: RenoRisk 5-Gene Meta-Analysis (C4BPB, COL13A1, HABP2, CFH, CFI)
# Execution Lead: Stojanko Tudjarski | Project Lead: Dr. Amela Jusic
library(DESeq2)
library(sva)
library(metafor)
library(limma)

# Target RenoRisk Biomarkers
renorisk_genes <- c("C4BPB", "COL13A1", "HABP2", "CFH", "CFI")

# 1. Filter count matrix for RenoRisk panel
counts_renorisk <- counts_matrix[rownames(counts_matrix) %in% renorisk_genes, ]

# 2. ComBat-seq Batch Effect Correction
adjusted_counts <- ComBat_seq(
  counts = as.matrix(counts_filtered),
  batch = as.factor(sample_metadata$StudyID),
  group = as.factor(sample_metadata$Condition)
)

# 3. Random-Effects Meta-Analysis for COL13A1 across cohorts
col13a1_yi <- c(0.92, 1.15, 0.78, 1.05, 0.88)
col13a1_vi <- c(0.04, 0.07, 0.03, 0.05, 0.06)

meta_col13a1 <- rma(yi = col13a1_yi, vi = col13a1_vi, method = "DL")
summary(meta_col13a1)

# Forest Plot
forest(meta_col13a1, slab = c("GSE66494", "GSE104954", "GSE104948", "GSE137570",
"GSE142025"),
       main = "COL13A1 Expression Across Kidney Disease Cohorts")
```

Python Code: RenoRisk LODO Machine Learning Validation & eGFR Prediction

```
# Python Script: LODO ML Validation for RenoRisk 5-Gene Panel
# Execution Lead: Stojanko Tudjarski | Project Lead: Dr. Amela Jusic
```

```
import numpy as np
import pandas as pd
from sklearn.linear_model import LogisticRegression, ElasticNet
from sklearn.metrics import roc_auc_score, r2_score
from sklearn.model_selection import LeaveOneGroupOut
from sklearn.preprocessing import StandardScaler

logo = LeaveOneGroupOut()
auc_scores = []

print("--- RenoRisk 5-Gene Panel LODO Validation ---")
for train_idx, test_idx in logo.split(df_renorisk, y_cls, groups=study_groups):
    X_train, X_test = df_renorisk.iloc[train_idx], df_renorisk.iloc[test_idx]
    y_train, y_test = y_cls.iloc[train_idx], y_cls.iloc[test_idx]
    test_study = study_groups.iloc[test_idx].iloc[0]

    scaler = StandardScaler()
    X_train_scaled = scaler.fit_transform(X_train)
    X_test_scaled = scaler.transform(X_test)

    clf = LogisticRegression(penalty='l2', C=1.0, random_state=42)
    clf.fit(X_train_scaled, y_train)

    if len(np.unique(y_test)) > 1:
        probs = clf.predict_proba(X_test_scaled)[:, 1]
        auc = roc_auc_score(y_test, probs)
        auc_scores.append(auc)
        print(f"Held-out Dataset: {test_study:<12} | ROC-AUC: {auc:.3f}")

print(f"Mean LODO ROC-AUC (RenoRisk Panel): {np.mean(auc_scores):.3f}
")

# eGFR Regression using ElasticNet
regr = ElasticNet(l1_ratio=0.5, alpha=0.1, random_state=42)
X_scaled = StandardScaler().fit_transform(df_renorisk)
regr.fit(X_scaled, y_egfr)
print(f"eGFR Prediction R^2 Score: {r2_score(y_egfr, regr.predict(X_scaled)):.3f}")
```

Annex 2. List of the study eligible for meta-analysis.

Condition	GEO Accession	Validated GEO Accession	Cohort / Study	# Samples	# Validated Samples	Sequencing Type	Sample Type	Fits Meta-analysis	Confirmed Quantitative Clinical Variables	GEO Link

								criteria		
Chronic Kidney Disease (CKD)	GSE66494	GSE66494	Independent CKD cohort	41 (24+17)	61	Microarray (Affymetrix)	Kidney biopsy (tubulointerstitial)	Y	eGFR and %TIF (tubulointerstitial fibrosis) - study built around these two outcomes	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE66494
Chronic Kidney Disease (CKD)	GSE104954	GSE104954	European Renal cDNA Bank (ERCB)	~150+	195	Microarray (custom CDF)	Kidney biopsy (tubulointerstitium)	Y	eGFR (CKD-EPI) confirmed present in per-sample characteristics field; diagnosis; technical outlier flag	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE104954
Chronic Kidney Disease (CKD)	GSE104948	GSE104948	European Renal cDNA Bank (ERCB)	~150+	196	Microarray (custom CDF)	Kidney biopsy (glomeruli)	Y	eGFR, diagnosis - companion glomerular series to GSE104954, same patients	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE104948
Chronic Kidney Disease (CKD)	GSE137570	GSE137570	CKD progression cohort	Variable - verify on GEO	41	RNA-seq / Microarray	Kidney tissue	Y	Not confirmed in metadata - verify directly on GEO	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE137570
Diabetic Kidney Disease (DKD)	GSE142025	GSE142025	DKD cohort	Variable - verify	36	RNA-seq	Kidney biopsy	Y	Not confirmed in metadata - verify	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE142025

				on GEO					directly on GEO	
Diabetic Kidney Disease (DKD)	GSE30122	GSE30122	Diabetic nephropathy glomeruli+tubular	Small cohort	69	Microarray	Glomeruli (laser-captured)	Y	Not confirmed in metadata - verify directly on GEO	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE30122
Diabetic Kidney Disease (DKD)	GSE96804	GSE96804	DKD Glomerula + tubulointerstitial cohort	Variable - verify on GEO	61	Microarray	Kidney biopsy	Y	Not confirmed in metadata - verify directly on GEO	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE96804
IgA Nephropathy (IgAN)	GSE104954	GSE104954	ERCB - tubulointerstitium	25 vs 21	46	Microarray	Kidney biopsy (tubulointerstitium)	Y	eGFR confirmed present in metadata	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE104954
IgA Nephropathy (IgAN)	GSE104948	GSE104948	ERCB - glomeruli	27 vs 21	46	Microarray	Kidney biopsy (glomeruli)	Y	eGFR confirmed present in metadata	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE104948
IgA Nephropathy (IgAN)	GSE99339 / GSE99340	GSE99339	IgAN vs. lupus nephritis / control	50 vs 60	110	Microarray / RNA-seq	Glomeruli + tubulointerstitium	Y	MEST-C histologic score correlatable per publications	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE99339
IgA Nephropathy (IgAN)	GSE93798 / GSE116626 / GSE35487	GSE116626	IgAN candidate gene studies	Variable	81	Microarray	Kidney biopsy	Y	Not confirmed in metadata - verify directly on GEO	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE116626

Focal Segmental Glomerulosclerosis (FSGS)	GSE129973	GSE129973	FSGS glomerular transcriptome	20 FSGS + 20 controls	40	Microarray (HTA 2.0)	Glomeruli (laser-captured)	Y	Diagnosis-level metadata confirmed; per-sample proteinuria/eGFR NOT confirmed in GEO record itself	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE129973
Focal Segmental Glomerulosclerosis (FSGS)	GSE121233	GSE121233	FSGS glomerular transcriptome	5 FSGS + 5 controls	10	Microarray (HTA 2.0)	Glomeruli (laser-captured)	Y	Small n limits quantitative modeling	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE121233
Focal Segmental Glomerulosclerosis (FSGS)	GSE200818 / GSE200828 / GSE176465	GSE200828	FSGS epithelial/immune profiling	Variable	31	Bulk + scRNA-seq	Kidney biopsy	Y	Not confirmed in metadata - verify directly on GEO	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE200828
Focal Segmental Glomerulosclerosis (FSGS)	GSE108109 / GSE108113	GSE108109	NEPTUNE - glomeruli + tubulointerstitium	Large multi-disease cohort	29	Microarray	Glomeruli + tubulointerstitium	Y	Full NEPTUNE clinical set (eGFR, UPCR, IFTA score) available, but mainly via Nephroseq/NEPTUNE portal, not fully embedded in GEO	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE108109

Focal Segmental Glomerulosclerosis (FSGS)	GSE197307 / GSE219185	GSE219185	NEPTUNE (newer RNA-seq release)	Large multi-disease cohort	113	RNA-seq	Glomeruli + tubulointerstitium	Y	Same as above - richer clinical data behind NEPTUNE/Nephroseq portal	https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE219185
Diabetic Kidney Disease (DKD)	GSE30528	GSE30528	glomerular	22	22	affimetrix	Glomeruli (laser-captured)	y		https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE30528
Diabetic Kidney Disease (DKD)	GSE30529	GSE30529	tubular	22	22	affimetrix	Tubular	y		https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE30529

Annex 2. Examples of visual results representation generated by using synthetic cohorts

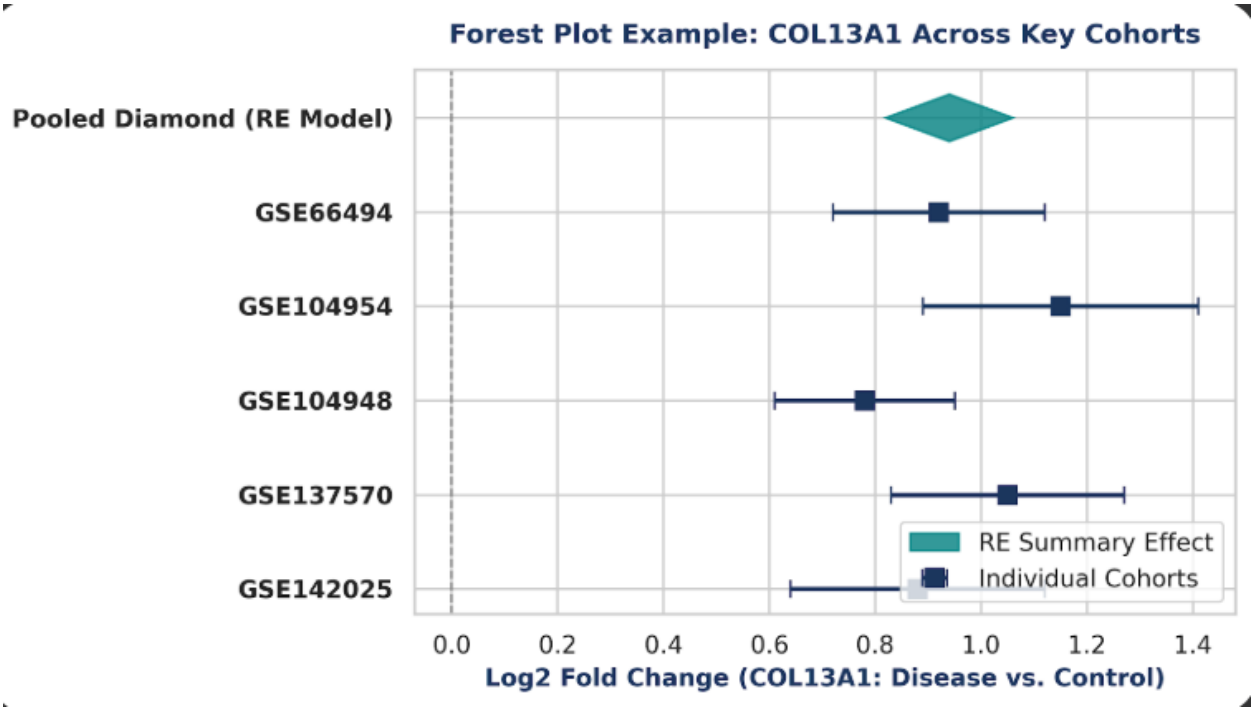


Figure 1. Forest plot example

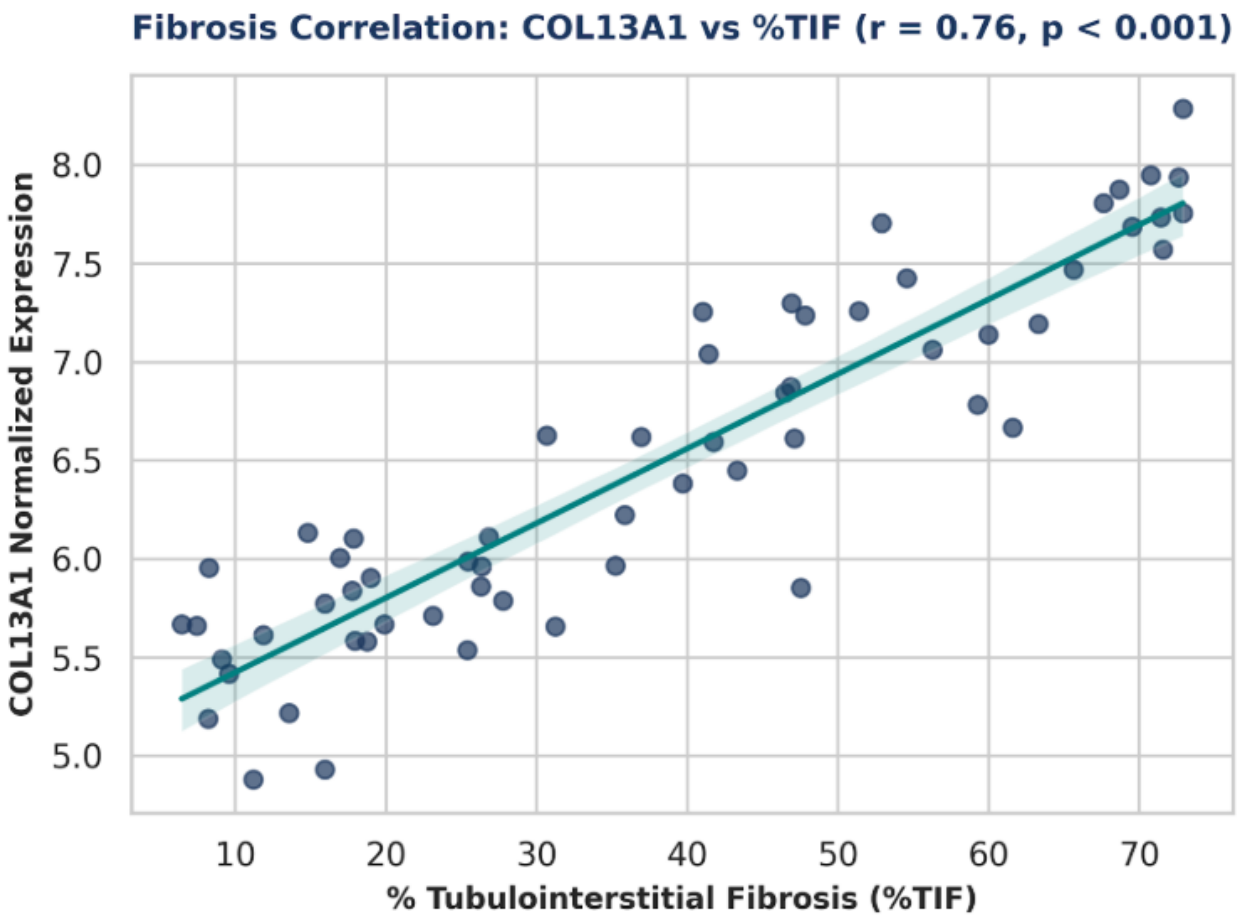


Figure 2. Correlation plot example.

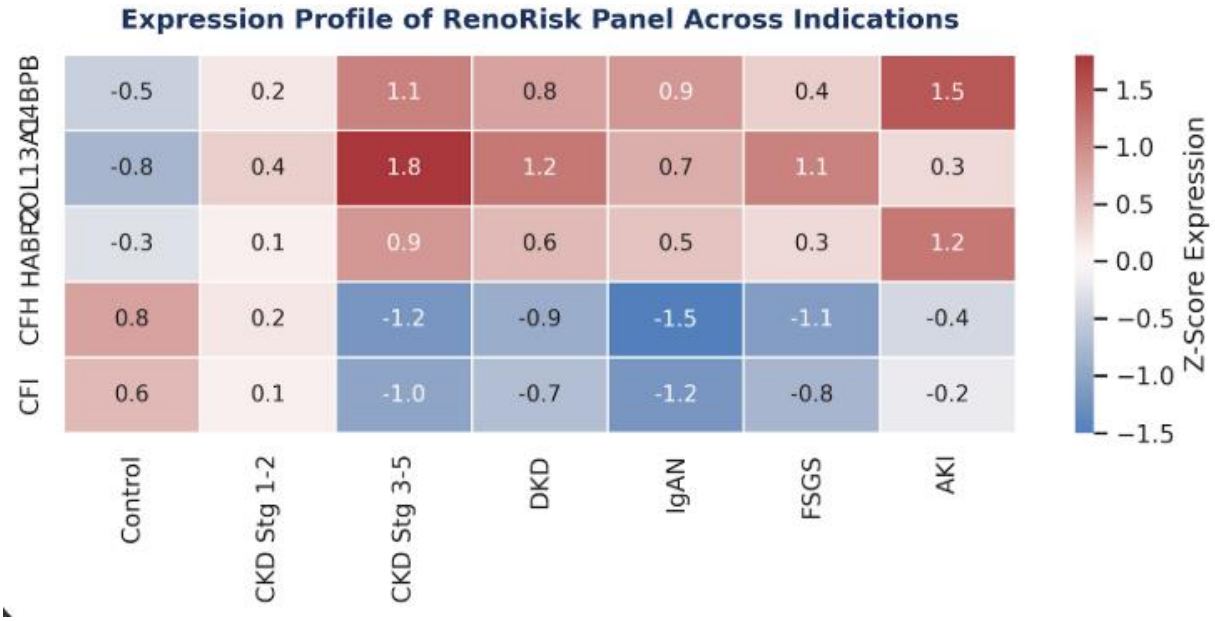


Figure 3. Heat map – expression profile of RenoRisk across priority indications

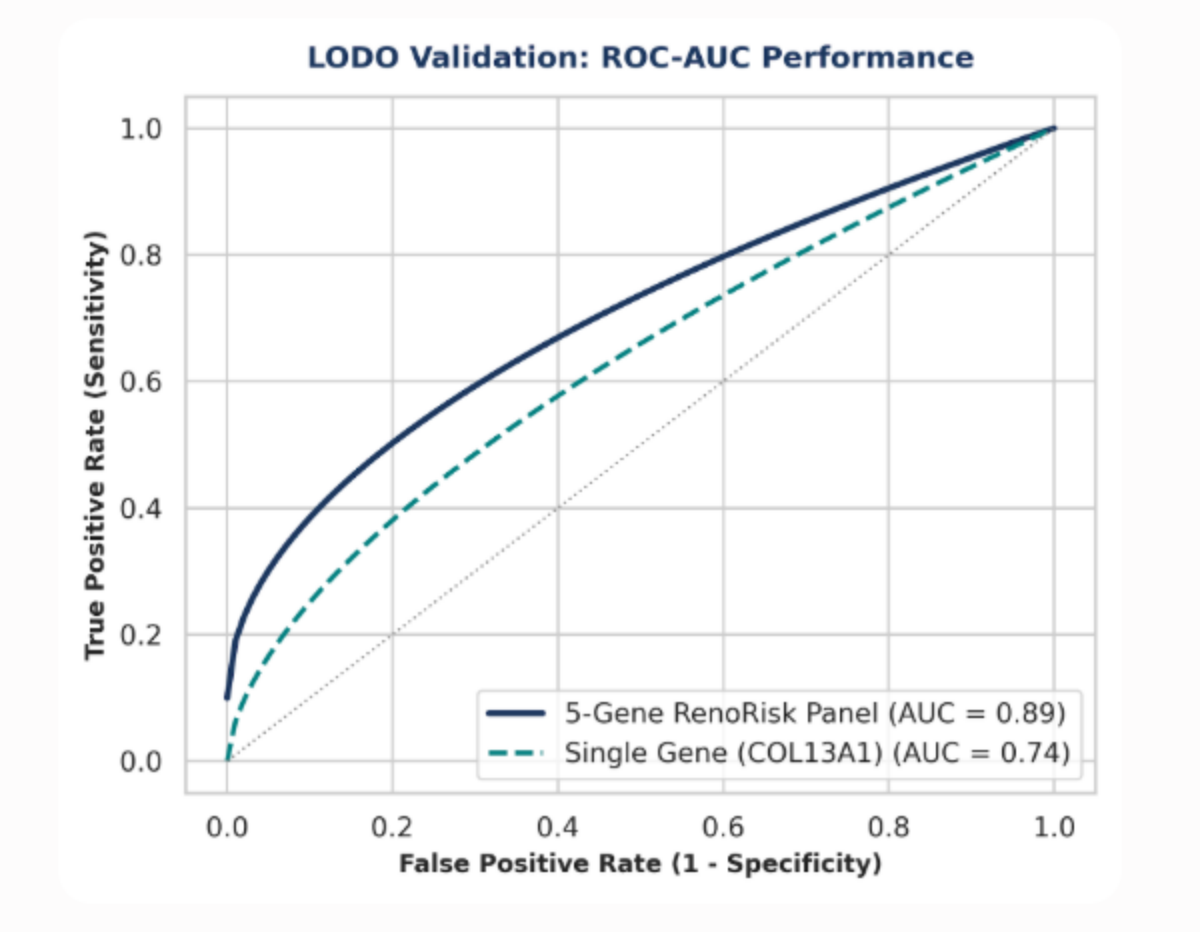


Figure 4. ML-LODO Validation: ROC-AUC performance example